

Reinforcement Learning and Dynamic Optimization – Final Project 2: Poker

Phase 1: LeDuc Poker

Team 17: Horn Pavlos, Vogiatzis Giorgos

In this phase, we implement our environment of the mini-poker variation called LeDuc: Hold'em. We did not take RLcard's environment setup, and we initially wrote our code based on the following documentation: [link](#)

We later adjusted the environment according to the professor's specifications, but some things may have still slipped. We trained several agents against fixed types of opponents. Specifically:

```
opp_styles = ['Pal-Bot', 'Random', 'Tight', 'Limping', 'Aggressive',  
'Honest-Strength', 'Mixed']
```

Mixed opponents are opponents who shift between the first 6 styles uniformly between testing episodes. Pal-Bot is a custom fixed opponent inspired by the behavior of our dear friend Mitsos Palaiologos. It aims to portray an aggressive opponent who bluffs more smartly (e.g. when the bot has a 'JJ' pair, it tries to bait the tabular Q-learning agent into raising). Code for the Q-learning agents is in `Poker_Agent.py`. Fixed opponent implementations are provided in `Player_Baselines.py`. The experiments lie in the .ipynb notebook, along with data and graphs stored in the subfolders of the project. Here, we provide the mean round rewards per trained agent and opponent style contained in the `Final_Results.csv` file. Format is a 6x7 matrix where rows represent the trained agents and columns represent the fixed opponents. Reward is measured in terms of big blinds (chips x 2) per round:

	Pal-Bot	Random	Tight	Limping	Aggressive	Honest-Strength	Mixed
Pal-Bot	0.21	0.17	-0.01	0.12	-0.44	0.04	0.01
Random	0.06	1.17	0.12	0.33	0.95	0.15	0.17
Tight	0.40	0.34	0.40	0.47	0.14	0.37	0.39
Limping	-0.03	0.43	-0.01	0.33	-0.16	0.04	0.05
Aggressive	0.08	1.05	0.15	0.42	0.80	0.18	0.23
Honest-Strength	0.15	0.40	0.30	0.41	0.15	0.35	0.35

We see that there is a problem with our implementation. While the agents perform positively when playing against opponents they trained for, they often seem to be sub-optimal against them. For example, the Pal-Bot trained agent yields an average of 0.21 big blinds per round against a Pal-Bot opponent, while a Tight trained agent yields a higher 0.40 against the same opponent. This need to be fixed.